

```
1 function [pcLin, r] = PulseCompression(Data, Config, WaterProp)
2
3     N=length(Data.RawEcho1_90kHz_SampleRate);
4     Z=Data.RawEcho1_90kHz_DataI+j*Data.RawEcho1_90kHz_DataQ;
5     Xmit=Data.RawEcho1_90kHzTx_DataI+j*Data.RawEcho1_90kHzTx_DataQ;
6     pcLin=zeros(N,length(real(Z(1,:)))-length(Xmit));
7     pcLinMat=zeros(N,length(real(Z(1,:)))+length(real(Xmit))-1);
8     for i=1:N
9         % pcLin(i,:) = abs(complexConv(Z(i,:),Xmit));
10        pcLinMat(i,:) = abs(conv(Z(i,:),Xmit));
11    end
12    pcLin = pcLinMat(:,[length(real(Xmit)):(end-length(real(Xmit)))]);
13    r=0:(length(pcLin(1,:))-1);
14    %r=r*750/Config.echo_rawSampleRate1;
15    r = r*(WaterProp.soundVel*0.5)/Config.echo_rawSampleRate1;
16 return
17
18
19
```